

Review Article

The Use of β -Adrenergic Receptor Antagonists in Psychiatry: A Review



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Background: β -Blockers (BBs) are a widely used class of medications with a number of medical indications. Their ability to suppress the adrenergic response has made them a target of investigation for treating a variety of psychiatric symptoms. **Objective:** The aim of this review is to identify evidence regarding the use of BBs for various psychiatric disorders including anxiety disorders, mood disorders, psychosis, and posttraumatic stress disorder. We also review the evidence for use for aggression and agitation and for extrapyramidal side effects of psychiatric medications. **Methods:** A search of PubMed and Google Scholar was conducted with search terms relevant to the use of BBs for psychiatric conditions. Studies were selected based on relevance to the review objective, focusing on systematic reviews and meta-analyses. Case reports were included and are clearly identified when referenced in this review. **Results:** A total of 44 publications were included that examined the use of BBs as treatment for the following psychiatric symptoms

and conditions: anxiety disorders, depression and mood disorders, acute stress disorder and posttraumatic stress disorder, psychotic disorders and extrapyramidal symptoms, as well as aggression and agitation. Additionally, adverse effects and drug-drug interactions were reviewed. **Conclusion:** Well-characterized psychiatric uses of BBs include treating symptoms of performance anxiety and akathisia. Some positive evidence exists for BB therapy in other disorders such as specific phobia, panic disorder, and for aggression in patients with psychosis, acquired brain injury, or intellectual disability. Most of the evidence on the role of BBs in psychiatry is as adjunctive treatment for management of somatic symptoms rather than as monotherapy. Outside of performance anxiety and akathisia, the role of BBs in psychiatry has not been subjected to careful study, and further investigative trials are needed.

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Key words: beta-blocker, adrenergic blockade, psychopharmacology, consultation liaison psychiatry.

INTRODUCTION

After the development of β -adrenergic receptor (β AR) antagonists in the 1950s and their initial use in cardiac disease, the variety of clinical applications of β -blockers (BBs) has increased greatly. BBs block the action of endogenous catecholamines, epinephrine, and norepinephrine, at β ARs located throughout the peripheral nervous system and central nervous system (CNS) as part of the autonomic nervous system. Studies have shown that many psychiatric disorders are affected or precipitated by increased release of catecholamines in

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both the peripheral and CNS particularly under conditions of repeated or uncontrolled stress.¹ This interplay between the autonomic nervous system and psychiatric disorders has led to expanding research and clinical applications for the use of BBs for various psychiatric indications. This review will assess the current literature on the clinical utility of β -adrenergic blockade for various psychiatric indications.

Understanding the psychiatric applications of BBs requires an understanding of β AR physiology. There are 3 known β AR subtypes designated as β_1 (heart and kidneys), β_2 (lungs and vasculature), and β_3 (fat cells). All 3 β ARs are expressed in the CNS to some extent and are coupled with G proteins that are linked to adenylate cyclase involved with mediating intracellular events after hormone binding.^{1,2}

Differences in binding affinities among BBs influence pharmacological action depending on their receptor profile (Table 1). Nonselective BBs demonstrate antagonism at both β_1 and β_2 receptors while cardioselective agents favor β_1 -adrenergic receptors.³ As a result, the β_1 -blocking activity of cardioselective BBs is particularly pertinent to heart disease. In the peripheral nervous system, catecholamines increase during periods of stress and result in increased heart rate (β_1), enhanced pulmonary function (β_2), and increased blood pressure (α_1) as well as the effects on the CNS through moderating norepinephrine action along projections throughout the brain.¹ The ability for BBs to affect the CNS depends on the ability to penetrate the blood-brain barrier and bind directly with β ARs as a result of their lipophilicity (Table 1). Therefore, lipophilic BBs, such as propranolol and metoprolol, more readily

cross through the blood brain barrier and have been reported to exhibit greater neuropsychiatric effects.⁵ Because of these effects, BBs have sometimes been used for management of psychiatric symptomatology, most often as an adjunct to treatment, or as a second line agent. This review summarizes evidence for the use of BBs for various psychiatric symptoms and disorders.

A literature search was conducted to review the evidence for the use of BBs for psychiatric disorders, using PubMed and Google Scholar with search terms including “beta blockers,” “adrenergic blockade,” “psychiatry,” “psychiatric disorders,” “anxiety,” “performance anxiety,” “panic disorder,” “generalized anxiety disorder,” “specific phobia,” “mood disorder,” “depression,” “bipolar disorder,” “mania,” “post-traumatic stress disorder,” “acute stress disorder,” “psychosis,” “schizophrenia,” “extrapyramidal symptoms,” “akathisia,” “agitation,” “aggression,” “brain injury,” and “intellectual disability.” In reviewing the search results, systematic reviews and meta-analyses were included to incorporate the most robust evidence-based literature available. Case reports were also included to illustrate salient points and are clearly identified when referenced.

ANXIETY DISORDERS

Performance Anxiety

Perhaps the best-known psychiatric application of BBs is in managing anxiety disorders, particularly the performance subtype of social anxiety. One of the earliest studies of using BBs for performance anxiety suggested that a single 40-mg dose of oxprenolol before public speaking reduced tachycardia and electrocardiographic abnormalities (including ectopic beats and J point depression).^{6,7} Since then, propranolol, atenolol, pindolol, and nadolol have all demonstrated efficacy in reducing the symptoms of performance anxiety. These agents include both selective and nonselective, as well as high- and low-lipophilicity BBs. This variety suggests that beneficial effects are achieved not through CNS activity as might be expected for lipophilic BBs such as propranolol, but rather due to the common pathway of suppressing the cardiovascular response to catecholamines. Unfortunately, there has been little investigation comparing the relative benefits of BBs with differing selectivity and lipophilicity. Regardless, these benefits appear to consist of symptomatic relief rather

TABLE 1. Characteristics of β -Blockers Discussed in This Review

Beta blocker	Receptor selectivity	Lipophilicity
Atenolol	β_1	Low
Betaxolol	β_1	Moderate
Carvedilol	$\beta_1, \beta_2, \alpha_1$	High
Labetalol	$\beta_1, \beta_2, \alpha_1$	High
Metoprolol	β_1	Moderate
Nadolol	β_1, β_2	Low
Oxprenolol	β_1, β_2	Moderate
Pindolol	β_1, β_2	Moderate
Propranolol	β_1, β_2	High
Timolol	β_1, β_2	Moderate

Lipophilicity: low = LogP <1, moderate = LogP 1–3, high = LogP >3.^{3,4}

than long-term treatment of an underlying psychiatric condition. A limiting factor of these studies is that many investigations of using BBs for social anxiety disorder have not used a standard screening tool for diagnosis.^{8,9}

A Cochrane review of pharmacotherapy for social anxiety disorder found a “relative lack of empirical support” for using BBs for performance anxiety because of the exclusion of many studies failing to clearly establish a diagnosis of social anxiety disorder in their subjects.⁸ Despite the lack of an established role for BBs in “rigorously diagnosed” social anxiety disorders, a literature review cites multiple studies demonstrating improvement in performance-related anxiety symptoms and technical performance among subjects who were given BBs before performance.⁹ Multiple studies have shown pretreatment with BBs enhances technical performance in musicians performing in public, likely by controlling somatic symptoms such as tremor, and for this purpose, it may outperform benzodiazepines.^{10–12} BBs are preferable because of their lower potential for side effects and misuse than other medications used for performance anxiety.¹⁰

Panic Disorder

Unlike performance anxiety, panic attacks are more difficult to treat acutely with BBs because of their unpredictable onset and short duration.¹⁰ One small, double-blind, placebo-controlled, cross-over study found that 40 mg of propranolol given just before a medically induced panic attack (via carbon dioxide inhalation) reduced tachycardia but had little effect on subjective anxiety.¹³ Small studies of daily prophylactic BBs showed some efficacy in reducing somatic symptoms and number of panic attacks, particularly when used concurrently with selective serotonin reuptake inhibitors for otherwise refractory panic disorder, but their effect on subjective anxiety and avoidance behavior is less consistent.^{10,14–16} Notably, these studies included both cardioselective and nonselective BBs. While it might be expected that lipophilic agents that can cross into the CNS would more effectively control subjective and behavioral symptoms, the limited data provided by these investigations are insufficient to demonstrate any BB subtype as obviously superior at controlling any particular symptom. A meta-analysis found insufficient evidence to support routine use of propranolol in panic disorder. However, the

randomized, blinded trials that were evaluated in this review found no statistically significant differences between the efficacy of propranolol and benzodiazepines regarding the short-term treatment of panic disorder with or without agoraphobia.¹⁷ BBs may have some benefit in augmenting standard pharmacotherapy, but there is a lack of randomized controlled trials (RCTs) comparing BB monotherapy to placebo.¹⁸

Generalized Anxiety Disorder

There have been very few studies on the use of BBs for generalized anxiety disorder.¹⁹ Early studies showed that while somatic generalized anxiety disorder symptoms (e.g., tachycardia) improve with BB therapy, benefits to subjective anxiety are inconsistent.²⁰ An open-label trial including 31 patients with generalized anxiety disorder found that a long-acting BB, betaxolol (doses ranging from 5 mg daily to 20 mg twice daily), was associated with improvement in generalized anxiety disorder severity.¹⁴

Specific Phobias

Relative to other anxiety disorders, there has been limited investigation into pharmacologic therapies for specific phobias, and exposure therapy remains the mainstay of treatment.⁶ However, a few small studies showed benefit from using BBs in some specific phobias. A study of 45 patients with fear of flying showed improvement in somatic and subjective symptoms compared with placebo after giving 100 mg of atenolol 3 times over a 16-hour period before a flight.²¹ Subjects with a phobia of dental appointments who were given 80 to 120 mg of propranolol one hour before dental procedures reported lower subjective anxiety than subjects given placebo.²² Atenolol is a cardioselective, hydrophilic BB, whereas propranolol is nonselective and highly lipophilic, suggesting that the somatic and subjective benefits perceived are largely mediated through peripheral, cardiovascular effects. Overall, responsiveness is inconsistent across different BBs and specific phobias, and it is even possible that BBs may blunt the efficacy of exposure therapy.^{23–25} Nevertheless, BBs may have a role in specific phobias, particularly in predictable situations such as flying or dentist appointments, when a BB could easily be taken before exposure.

MOOD DISORDERS

Depression as an Adverse Effect

Our understanding of the relationship between BBs and depression has shifted over time. Case reports and observational studies have suggested that the use of BBs was associated with depression (with a particular link to nonselective, lipophilic agents such as propranolol, pindolol, and timolol); however, more recent studies have not demonstrated a causal relationship.^{26,27} An analysis of 15 RCTs totaling greater than 35,000 patients that reviewed adverse effects of BBs in relation to fatigue, depression, and sexual dysfunction and how they related to selectivity and lipid solubility found no significant increase in depressive symptoms among those receiving any BB therapy. There was a statistically significant risk of fatigue reported for nonselective BBs; however, no risks of adverse effects were seen when categorized by different lipid solubility.²⁶ A more recent multicenter trial did not find a significant difference in depression symptoms when subjects were initiated on a BB after an acute myocardial infarction and compared to a control group.²⁸

Depression

Another study looked at patients without recent antidepressant use who suffered traumatic nonbrain injuries and found that those who had started a BB in the year preceding their injury were significantly less likely to require antidepressants in the 12 months after their injuries than those not receiving BBs.²⁹ The authors hypothesize that intense emotions experienced during a traumatic event contribute to the development of depression and that adrenergic blockade blunts emotional responses during such experiences.²⁹ This study is limited by its retrospective nature and the fact that it compared rates of antidepressant prescription in the year after injury as a surrogate for depression, rather than using diagnostic criteria.

A meta-analysis of 5 RCTs showed that high doses of pindolol (7.5 mg daily) as adjunctive treatment to selective serotonin reuptake inhibitors improved depressive symptom scores compared with lower doses of pindolol when given to patients with depression that had been refractory to treatment with selective serotonin reuptake inhibitors alone.³⁰ In addition, a recent population-based study of patients on antihypertensive medications which included over 700,000 subjects who

had received BB therapy found that propranolol, atenolol, bisoprolol, and carvedilol were associated with a decreased rate of diagnosis of depression relative to the comparison group of those only on diuretics for hypertension.³¹ These BBs include cardioselective and nonselective agents as well as high and low degrees of lipophilicity. The hazard ratios reported for these BBs were similar in magnitude, but no formal analysis was conducted to differentiate the effects based on BB characteristics. Apart from this recent study, much of the available literature addresses how BBs relate to depressive symptoms rather than using diagnostic criteria for major depressive disorder.

Mania

Several early studies investigated BB therapy for manic symptoms on the theoretical basis that the antiadrenergic effects of BBs might counteract the elevated, expansive mood and activity of mania.^{32,33} Some evidence suggested that BBs may improve manic symptoms; however, these findings were based on case reports and a small, uncontrolled study of 17 patients. The propranolol doses used in this study were very high (>1 g daily), theoretically increasing the risk of side effects (bradycardia, hypotension, bronchospasm), potentially limiting practical applications.³³ Further studies are needed before drawing conclusions regarding BBs as treatment for mania or bipolar depression.

ACUTE STRESS DISORDER AND POSTTRAUMATIC STRESS DISORDER

A few studies have evaluated using BBs for treating acute stress disorder and posttraumatic stress disorder (PTSD). It has been theorized that the development of these disorders relies on the neurocognitive concept of reconsolidation, in which a consolidated memory becomes increasingly durable as it is repeatedly recalled with intense emotions and, conversely, may lose potency if recalled with impaired reconsolidation.³⁴ The amygdala is thought to play an important role in creating the persistent, emotional memories of acute stress disorder and PTSD. Several studies have found that propranolol decreases amygdala activity and could disrupt the process of reconsolidation and prevent development of acute stress disorder and PTSD with initiation of propranolol soon after a traumatic

experience.^{35,36} Results have been inconsistent, and a meta-analysis and Cochrane review found insufficient evidence for using propranolol to prevent PTSD after traumatic events.^{17,37} Similarly, a meta-analysis showed that early administration of BBs within one month of a traumatic event did not prevent either acute stress disorder or PTSD.³⁸

In addition, individual RCTs failed to demonstrate a clinical benefit with even earlier administration of BBs, including one study that gave propranolol to trauma patients within 4 hours of their injury.³⁹

Other research has focused on diminishing the characteristic physiologic hyperarousal when a patient with PTSD is reliving a traumatic event. In 2 uncontrolled studies, propranolol appeared to blunt physiologic symptoms during script-driven re-exposure to traumatic memories and even seemed to have a durable effect lasting months after receiving the medication.⁴⁰ Several subsequent placebo-controlled studies, however, failed to demonstrate a similar effect.³⁴ A recent placebo-controlled study by Brunet et al. found that propranolol enhanced the effects of traumatic memory reactivation psychotherapy sessions but with high study attrition.⁴¹ This suggests that BBs may play only a limited role and specifically in augmenting trauma-focused therapies.

PSYCHOTIC DISORDERS

Extrapyramidal Symptoms

Pharmacologic management of psychotic disorders such as schizophrenia often requires medications with adverse side effects, such as extrapyramidal symptoms including akathisia. Although not indicated for other extrapyramidal symptoms, BBs are used as a first-line antiakathisia agent in clinical practice.^{42,43}

Propranolol is the most-studied BB for the treatment of akathisia. There are very limited data that metoprolol may provide comparable benefit for akathisia.⁴⁴ A single trial of nadolol found it is no better than placebo for treatment of akathisia.⁴⁵ Both nadolol and propranolol are nonselective. Nadolol has low lipophilicity (and consequently low CNS penetration) in contrast to the highly lipophilic propranolol and moderately lipophilic metoprolol. It is not clear, however, whether lipophilicity confers efficacy among BBs in the treatment of akathisia, or if receptor selectivity is a more relevant consideration.

Schizophrenia

BBs have also been investigated as an adjunct for improving symptoms in patients with schizophrenia as measured by various clinical scales (e.g., Brief Psychiatric Rating Scale, Clinical Global Impression, Present State Examination, Nurses' Observation Scale for Inpatient Evaluation).⁴⁶ Although some individual trials reported positive results, a Cochrane Review found that the data and methodologies failed to provide adequate evidence that BBs provide any benefit when used as adjunctive therapy for schizophrenia.⁴⁶ The authors also note that the research available is dated and has other substantial limitations.⁴⁶

AGGRESSION AND AGITATION

There is some evidence supporting the use of BBs as adjunctive treatment for acute agitation. Daily administration of pindolol (5 mg 3 times daily) in addition to conventional antipsychotic therapy decreased the number of aggressive incidents and overall psychiatric symptoms in schizophrenic patients with a history of aggressive behavior in a small, double-blinded, placebo-controlled trial.⁴⁷ Similar results were found with nadolol (120 mg daily) in hospitalized schizophrenia patients.⁴⁸ BBs may offer a reasonable treatment option for agitation given their relatively benign side effect profile compared with other medications used for agitation in psychotic individuals, but more high-quality studies are needed.

Several randomized, placebo-controlled trials in the 1980s and 1990s showed efficacy of BBs for treating agitation after acquired brain injury, defined as brain injury because of a nonprogressive, single insult other than stroke, including mechanisms such as trauma, anoxic injury, encephalitis, and toxic ingestion.⁴⁹ Most of the studies discontinued other standing antiagitation medications but allowed for treatment of acute episodes with additional medications on an as-needed basis. A 2006 Cochrane review assessing management of agitation in acquired brain injury patients found that daily administration of BBs, specifically propranolol and pindolol, was effective in reducing number of agitation episodes and/or agitation intensity. In addition, BBs were found to be useful for both agitation and aggression whether occurring early or late after injury.⁴⁹ However, the review noted that these studies were small (≤ 21 patients), often used large doses (greater

than the recommended maximum dose of 320 mg propranolol daily), and had limited long-term follow-up. More recent systematic reviews assessed the same 4 studies evaluated in the Cochrane review, and literature searches did not find any additional high-quality trials.^{50,51}

Despite the lack of more recent trials and the limitations of older ones, these systematic reviews found that BBs demonstrate some efficacy in controlling agitation in patients with brain injuries while causing few adverse effects.^{50,51} Propranolol has been considered by some as the gold standard of treatment for agitated brain-injured patients with concomitant hypertension.⁵⁰

Management of agitation and aggression is also an important aspect of caring for those with intellectual disabilities, particularly when they may cause harm to themselves. A systematic review found efficacy in the use of BBs for agitation in the intellectually disabled but also notes the quality of the studies available is poor, with no RCTs, and that there is heterogeneity of BB doses and usage of concurrent psychotropic medication, making it difficult to draw conclusions about BBs as monotherapy versus adjunctive treatment.⁵²

The vast majority of evidence for the possible role of BBs in agitated patients comes from trials evaluating nonselective BBs.^{46,47,49,51} Case reports and a small trial of nine patients suggest a possible benefit of cardioselective agents in agitated acquired brain injury patients, but further comparative RCTs are needed to evaluate the impact of BB selectivity on the treatment of aggression and agitation.⁵¹

ADVERSE EFFECTS AND CONTRAINDICATIONS

Generally, BBs are well tolerated. A large meta-analysis found that many side effects, including depression, syncope, postural hypotension, and fatigue, were no more common among patients taking BBs than those taking placebo.⁵³ Another review found no link to depression, although very small increases in fatigue and sexual dysfunction were identified.²⁶ Other significant adverse effects are largely related to the cardiac effects of BBs, including dizziness, bradycardia, and risk of decompensation in patients with heart disease. There is also concern for increased risk for hypoglycemia in diabetic patients because of loss of recognition of hypoglycemic

symptoms such as sweating and anxiety, although this effect has been called into question.⁵⁴

Because of the potential to worsen bronchospasm and act against the β -agonists often used for management of asthma, BBs are relatively contraindicated in patients with moderate to severe bronchospastic disease including asthma and chronic obstructive pulmonary disease. However, β_1 -selective blockers used in patients with mild to moderate disease are often tolerated, especially when asthma therapy is simultaneously increased, and a Cochrane review found the same is true in chronic obstructive pulmonary disease.^{55,56}

Cardiac contraindications include second- or third-degree heart block or a symptomatic bradycardia unless a pacemaker is present; recent diagnosis or exacerbation of congestive heart failure; or persistent hypotension with a systolic blood pressure less than 90 mmHg.⁵⁷ For patients with concern for underlying cardiac issues, consultation with a cardiologist is essential.

DRUG-DRUG INTERACTIONS

While BBs are generally well tolerated outside of absolute contraindications, there may be a risk of interactions with commonly prescribed psychotropics that could result in clinically significant adverse effects. These interactions occur through either altering drug metabolism, thereby changing drug levels in serum, or by creating additive or antagonistic effects.

Most BBs are metabolized by the liver through the cytochrome P450 system. Propranolol, carvedilol, nebivolol, timolol, and metoprolol are metabolized through CYP450 2D6. Fluoxetine, paroxetine, duloxetine, and bupropion are strong inhibitors of CYP450 2D6 that may increase the effects of the BBs listed.⁵⁸ Elevated serum levels of BBs may be associated with an increased risk of bradycardia and hypotension.⁵⁹

Orthostatic hypotension is a common side effect of treatment with tricyclic and other antidepressants, monoamine oxidase inhibitors, and antipsychotics because of their affinity for the α_1 adrenoceptor. The addition of a BB to one of these medications may further attenuate the compensatory rise in cardiac contractility and heart rate triggered by the α_1 blockade, thereby exaggerating postural blood pressure changes. This drug-drug interaction also occurs in patients treated with α_1 antagonists such as prazosin that is often used for nightmares in PTSD.⁶⁰

CONCLUSION

BBs have been used for a variety of psychiatric disorders and symptoms. Current review of the literature most strongly supports the use of BBs for management of symptoms of performance anxiety and akathisia associated with antipsychotic medications. There is some evidence to suggest the use of BBs in specific phobias, panic disorder, and aggression in patients with psychosis, acquired brain injury, and intellectual disability, most often as adjunctive treatment. Apart from symptoms of performance anxiety and akathisia, we did not find evidence for BBs as first-line treatment for any psychiatric condition. More research is needed, ideally with well-designed RCTs to better elucidate the efficacy, tolerability, and suggested dosage of BBs, whether alone or as adjunctive treatment in other psychiatric disorders.

In general, many of the perceived benefits of BBs in psychiatric conditions seem to be related to their ability to control somatic symptoms rather than a specific psychiatric effect in the CNS. BBs are well tolerated and generally have mild side effects at therapeutic doses making them relatively low risk interventions in patients without contraindications, especially if they have a medical comorbidity that may otherwise be treated with a BB.

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